Chapter 6 on Instrumentation

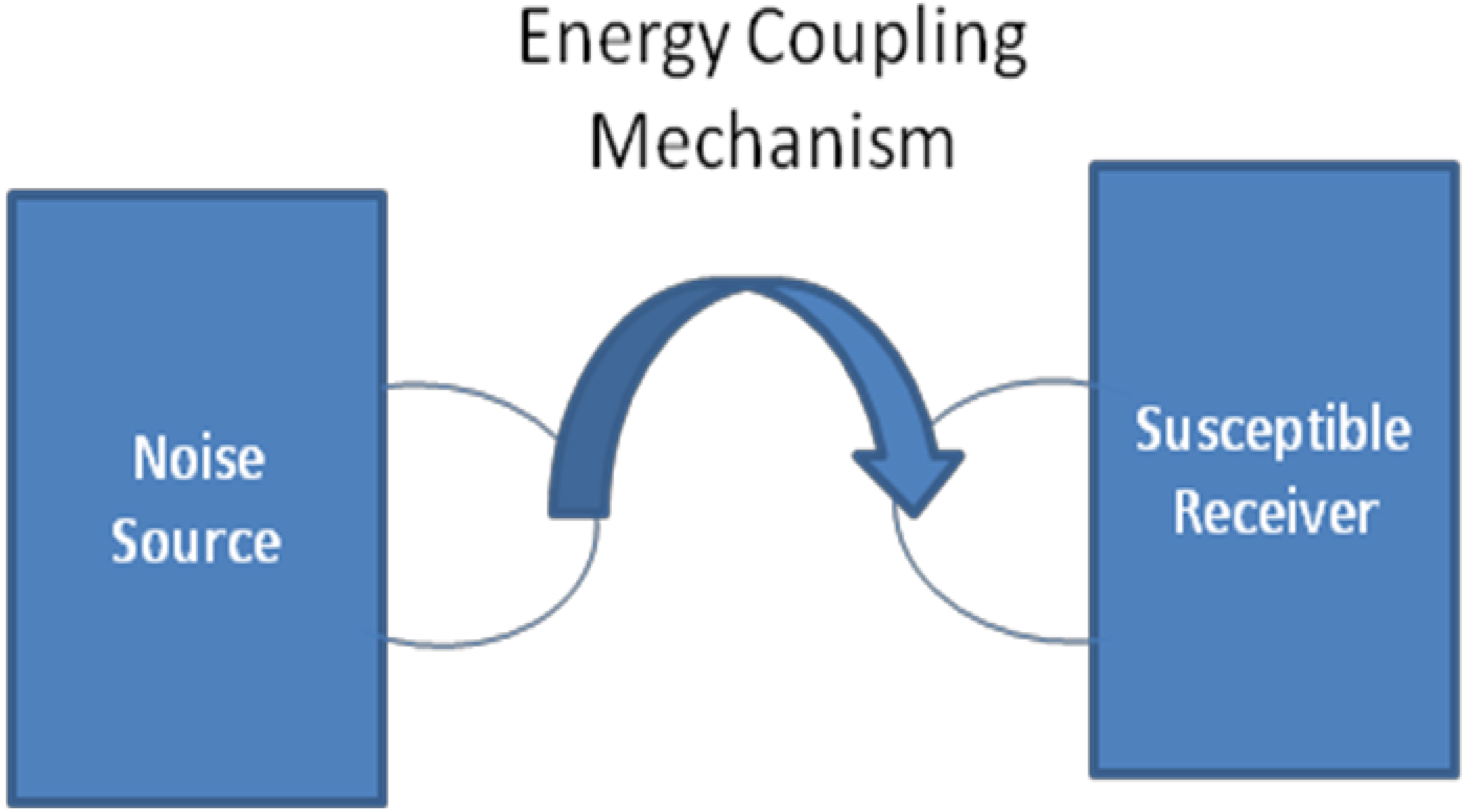
Unit 6: Circuit Design
(4 Hours / 4 Marks)

Noise, Noise (Energy) Coupling Mechanism and Prevention

- Noise is unwanted electrical activity coupled from one circuit into another.
- 3 components: A source, A coupling mechanism, and A receiver

Noise Sources

- Noise sources generate either a periodic signal or transient pulse that disrupts other circuits.
- There are many types of sources: Power lines, Motors, High voltage equipment (e.g. spark plug, igniter), Dischargers and sparks (e.g. lightning, static electricity), High current equipment (e.g. arc welder)


Energy Coupling
Mechanism
Noise
Source
Susceptible
Receiver

Energy Coupling Mechanism

Four mechanisms: Conductive, inductive, capacitive & electromagnetic

• Four mechanisms: Conductive, inductive, capacitive & electromagnetic

Coupling Mechanism	Frequency Range	Comment
Conductive	DC to 10 MHz	Requires a complete circuit loop (really no upper limit to frequency)
Inductive	Usually > 3KHz	Larger loop area in circuit means greater self inductance and mutual inductance associated with heavy current (can get significant coupling from 50 Hz-60 Hz power).
Capacitive	Usually > 1 KHz	Greater spacing between conductors reduces coupling associated with high voltage (can get significant coupling from 50 Hz-60 Hz power).
Electromagnetic	Usually > 15 MHz	Needs antenna s greater than 1/20 of wave length in both the source and susceptible circuit.

- Conductive coupling - low frequencies, caused by incorrect grounding.

- Capacitive & inductive coupling - dominate at high frequencies
  - Changing magnetic flux can couple circuits.
  - The loop area of the circuit is the primary factor that determines the inductance and coupling.
  - Changing electric potentials can drive charge through stray capacitances.
  - Appropriate grounding, shielding and signal separation control the amount of capacitive coupling.

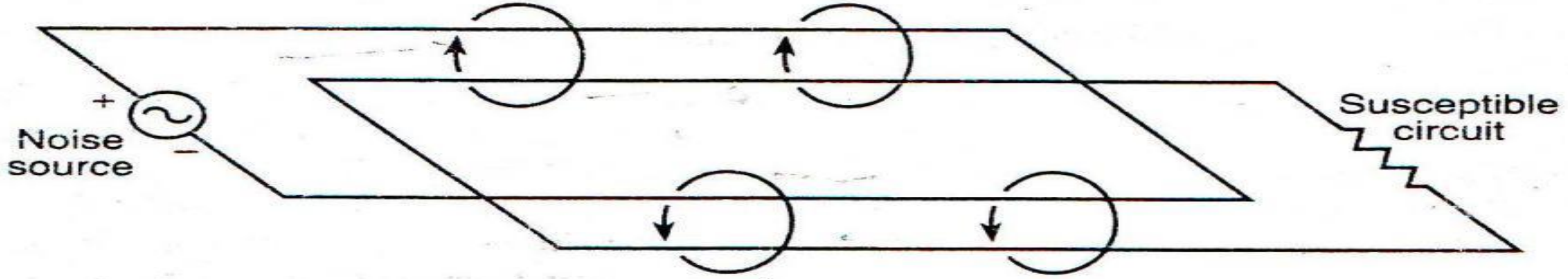


FIG. 6.6 Inductive coupling. Magnetic flux couples energy from the circuit of the noise source to the susceptible circuit.

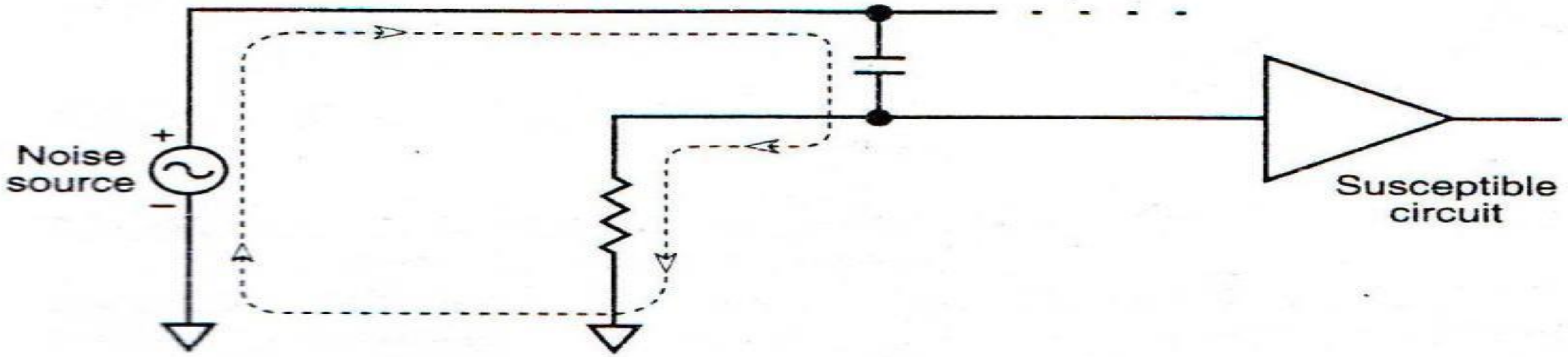


FIG. 6.7 Capacitive coupling. Coupling from stray capacitance completes the circuit of a noise source into a susceptible circuit.

- Electromagnetic coupling is a high frequency phenomenon.
- It requires a transmitting antenna in the source and a receiving antenna in the susceptible circuit. These antennas must be in appropriate fraction of the signal wavelength to couple effectively.

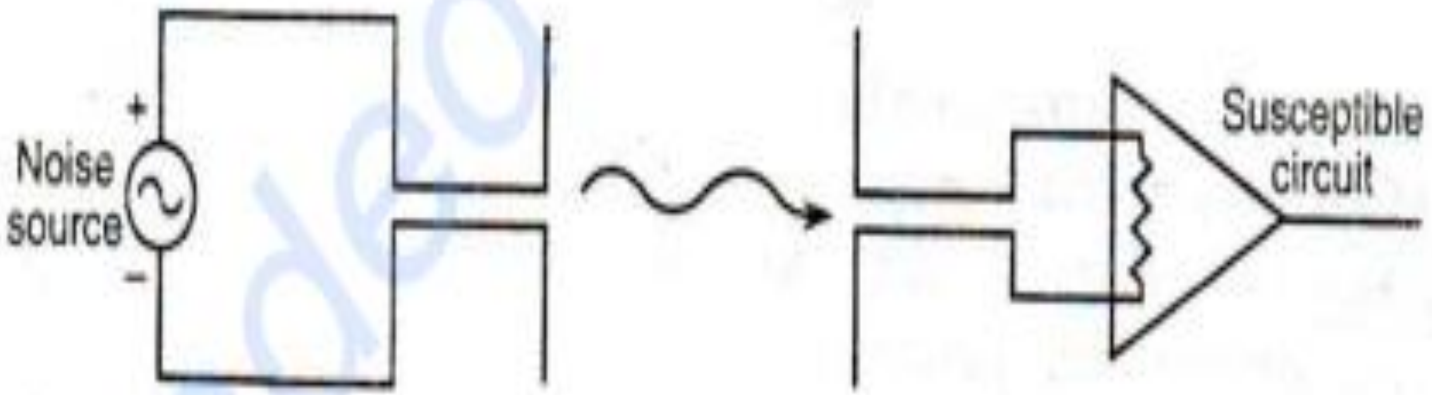


FIG. 6.8 Electromagnetic coupling. Radiated electromagnetic energy requires an antenna in both the noise and susceptible circuits. The antenna must be an appreciable portion of a wavelength, so such coupling is usually at high frequencies (> 15 MHz).

Susceptible Circuit

- Third component of noise is susceptible circuit.
- E.g. susceptibility includes cross talk on inputs that leads to bit flips in digital logic, radio interference and static discharge that destroy components.
- Susceptibility usually can be traced by proper grounding (or return paths) or long signal lines that are not properly shielded

Principle of Energy Coupling

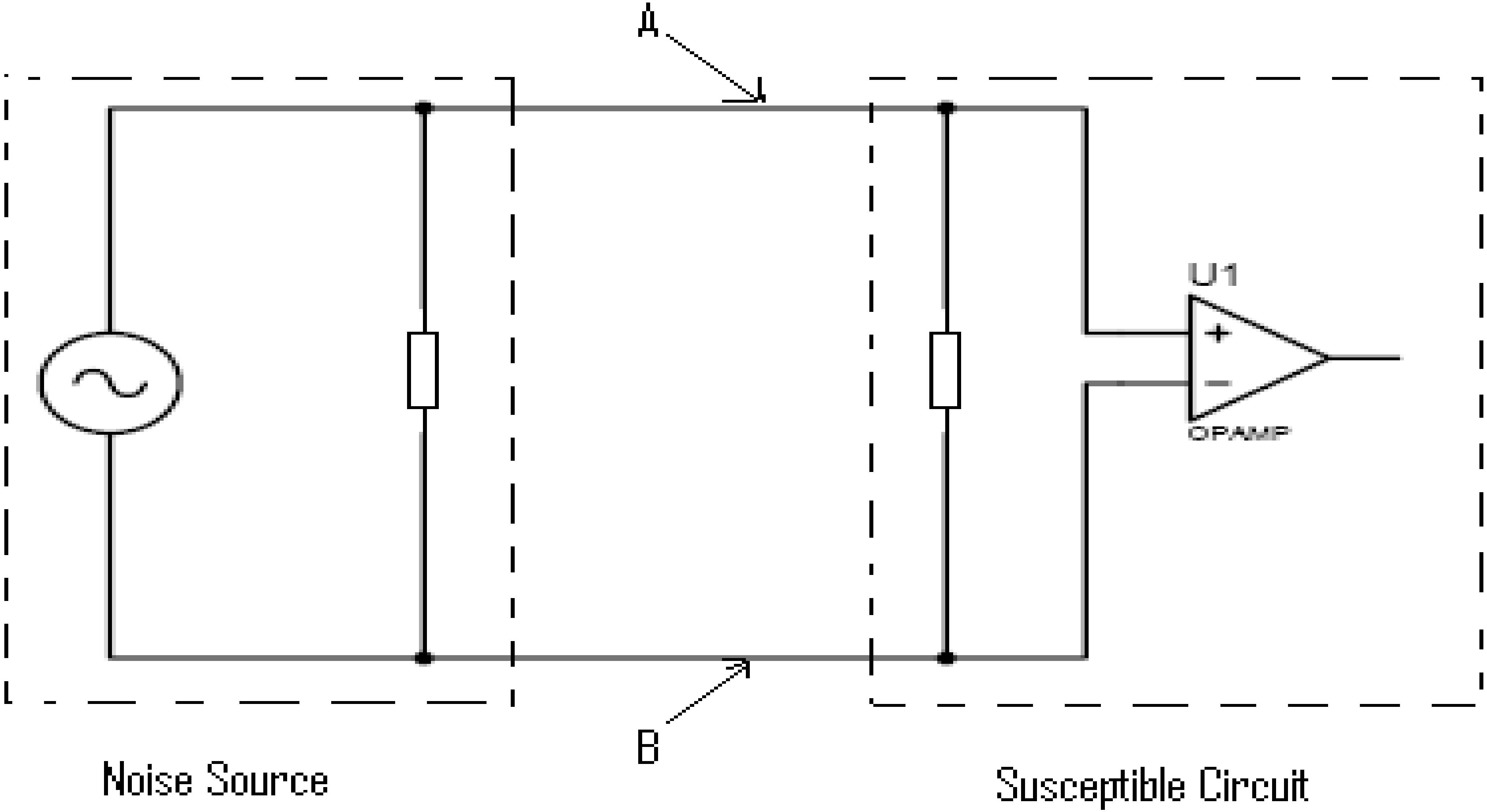
- Current will flow in the path with low impedance, not necessarily lowest resistance.
- Consequently, charge follows the path of minimum inductive and maximum capacitive reactance for the lowest impedance.
- $Z = SQRT(R^2+[WL - (1/WC)]^2)$

  Where, Z = Impedance, R = Resistance, WL = Inductive reactance & 1/WC = Capacitive reactance
- For frequencies above 3 KHz, a useful diagnostic for determining the mechanism is the ratio of rate of change in voltage to the rate of change in current.
- For the special cases of sinusoidal signals or resistive loads, the ratio is impedance; otherwise, it is a pseudo impedance value.
- A Diagnostic ratio called Pseudo Impedance.
- Pseudo Impedance is defined as: $\gamma = (dv/dt)/(di/dt)$
  - If $\gamma = 377$, @ high frequencies (> 20MHz) - Electromagnetic coupling.
  - f $\gamma < 377$, the value of di/dt > dv/dt i.e. large change in current - inductive coupling.
  - If $\gamma > 377$, the value of dv/dt > di/dt i.e. large change in voltage - capacitive coupling.

Conductive Coupling

- Requires a connection between source and receiver that completes a continuous circuit.
- Conductive coupling usually occurs at lower frequencies and is often caused by incorrect grounding.
- Such connections are inadvertent and difficult to find; such connections are called Sneak circuits.
- A ground loop is a complete circuit that allows unwanted current to flow into the ground.
- Substantial current in a ground path (as opposed to a return path) can produce voltage differences across the ground resistance and raise the ground potential at the loads.
- Conversely, significant potentials in the ground can force unwanted current to flow between circuits.
- The use of high frequency and reduction of ground loop can reduce conductive coupling or conductive noise.

Either connection A or B is removed, conductive noise is eliminated.



Inductive Coupling

An Inductive coupling mechanism requires a current loop that generates changing magnetic flux.

Generally, a current transient creates the changing magnetic flux, as follows:

Φ = BA = $\mu_0 n I A$

Then,

$$v = \frac{d\Phi}{dt} = A(\frac{dB}{dt}) = A\mu_0 n(\frac{di}{dt})$$

Where, Φ = Magnetic Flux

B = Magnetic field

A = Loop area

μ_0 = Permeability of free space

n = Number of turns in the loop

i = Current

v = Voltage

- The induced voltage in a magnetically coupled circuit is proportional to the time rate of change of current and loop area.
- Reducing the loop area will reduce the inductive reactance of a circuit.
- For frequencies above 3 MHz, (dv/dt) / (di/dt) $\ll 377\Omega$
- Generally, the load impedance is large, while the source impedance is small.
- Current follows the path of lowest impedance, not necessarily lowest resistance.
- Therefore, current will follow the path of minimum inductive reactance; this means the current will minimize loop area in a circuit.
- A slot in the ground plane of a circuit board will increase the loop area of a circuit; below figure shows this; so avoid such slots.

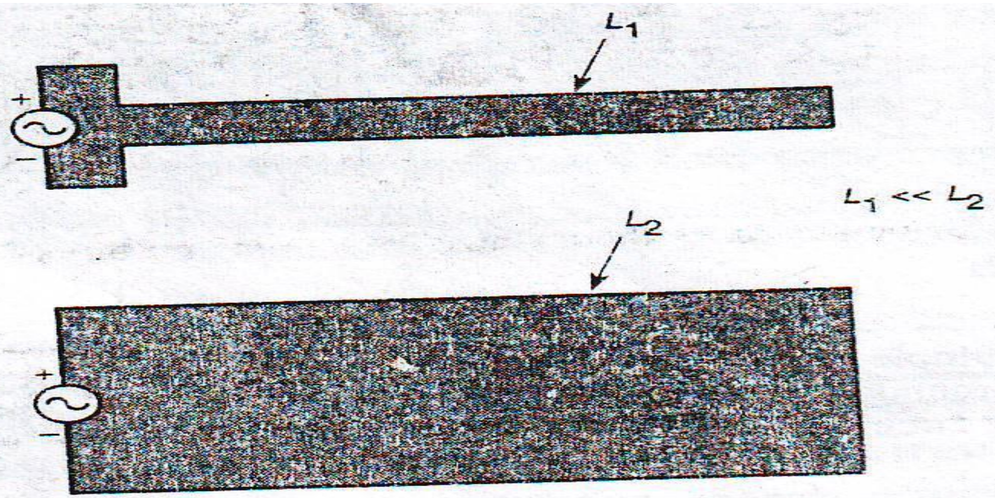


FIG. 6.11 Reducing loop area. Small loops of current have lower self-inductance and thus lower impedance.

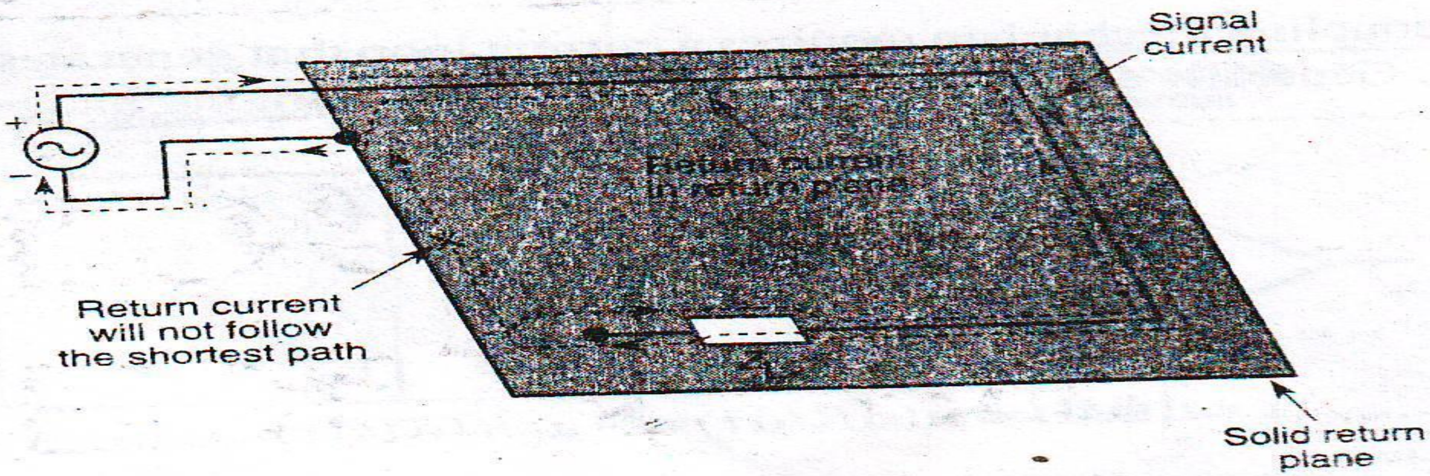


FIG. 6.12 Path of return current. Return current will follow the path of least impedance (in this case, the least self-inductance), not necessarily lowest resistance.

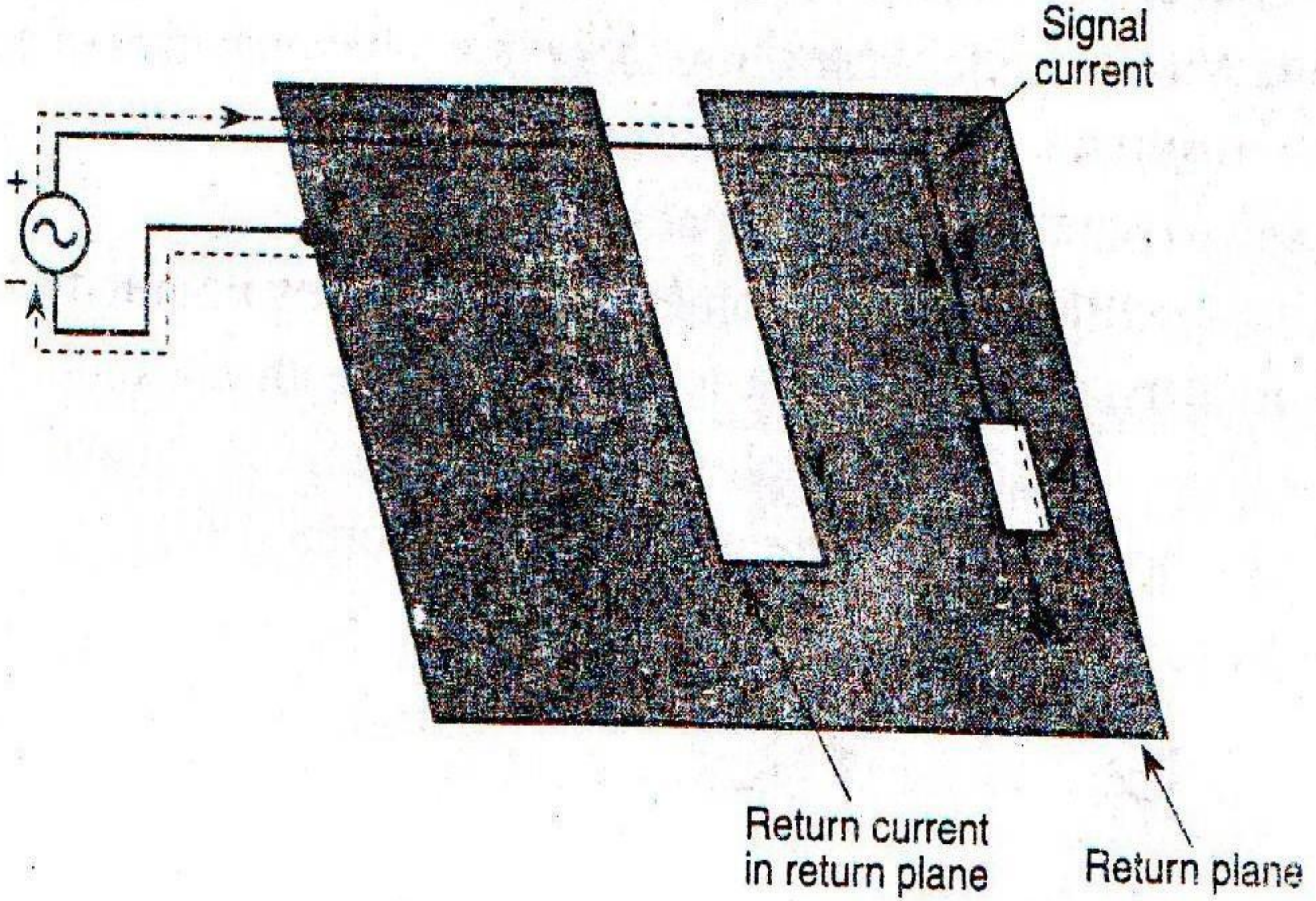


FIG. 6.13 A slot in the return plane increases the current loop area and the self-inductance.

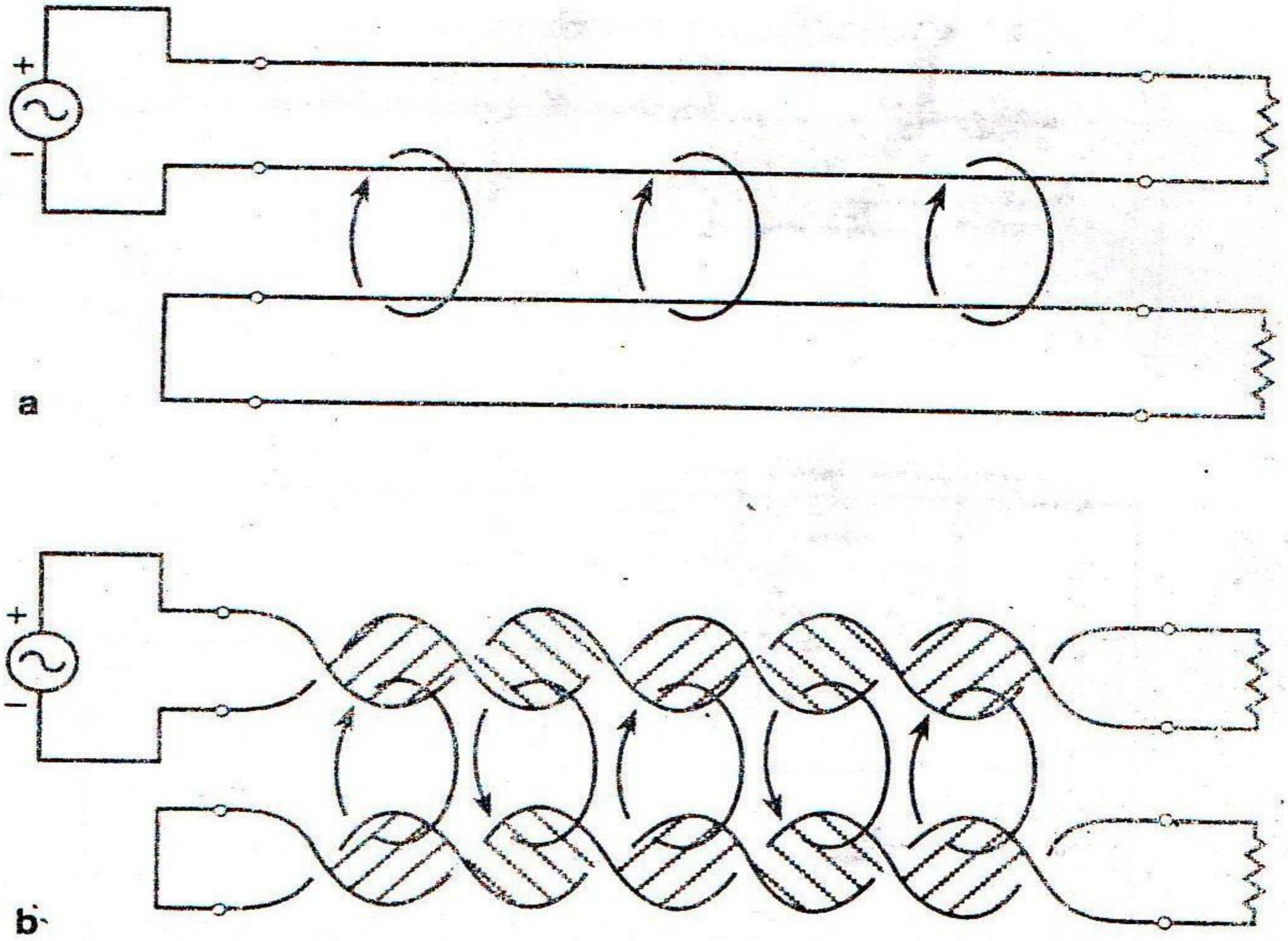


FIG. 6.14 Effect of twisted wire. (a) Straight wires create small loops that can couple magnetically. (b) Twisted wire eliminates the effective loop area of cables and magnetic coupling.

Capacitive Coupling

- Capacitive coupling mechanism requires both proximity between circuits and a changing voltage.
- It occurs when two conductors are placed at some distance apart and voltage level and frequency are changed.
- Capacitive coupling of noise becomes a factor for frequency above 1 KHz
- Generally, the total circuit impedance is high; i.e. both the source and load impedance are large.
- $(dv/dt) / (di/dt) >> 377\Omega$
- Capacitive coupling can be reduced by separation of conductors and appropriate shielding.

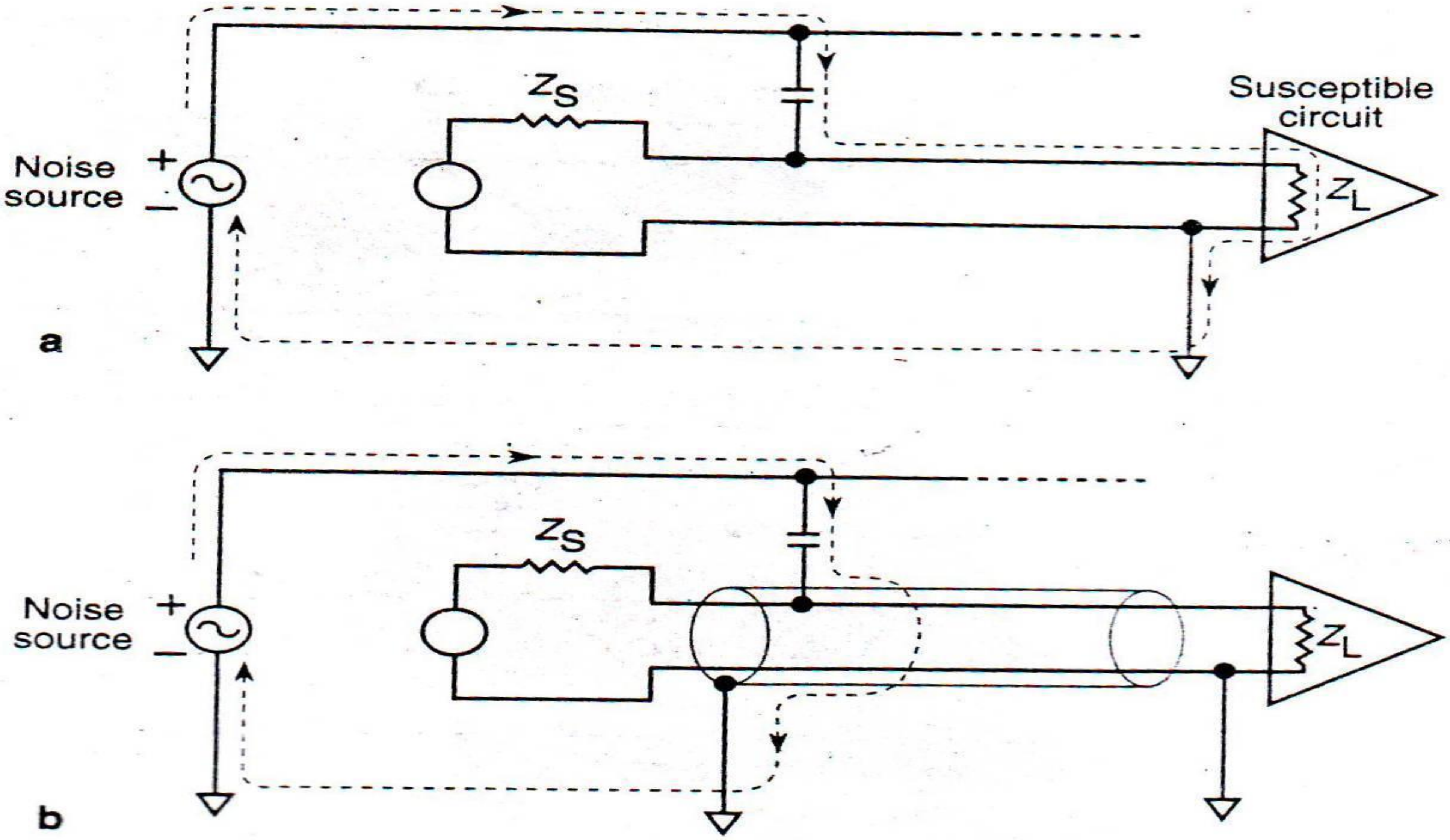


FIG. 6.15 Effect of shielding. (a) Without the shield, stray currents can disrupt susceptible circuits. (b) A properly connected shield can divert capacitively coupled current from susceptible circuits.

Electromagnetic Coupling

- Electromagnetic coupling or radiative coupling becomes a factor only when the frequency of operation exceeds 20 MHz .
- $f < 200$ MHz, cables are primary sources and receivers for electromagnetic coupling.
- $f > 200$ MHz, PCB traces begin to radiate & couple energy.
- Generally, the length of conductor must be longer than 5% of the wavelength, i.e., $l > \lambda/20$.
- Pseudo impedance factor between 100Ω and 500Ω.
- $(dv/dt) / (di/dt) \approx 377\Omega$
- The frequency of signal must be reduced.
- Use magnetic plate shielding.

Filtering

- Only filtering reduces conductive noise coupling.
- A filter can either block or pass energy by three criteria.

1. Frequency
    - LPF passes low frequency energy and rejects high frequency energy
    - HPF passes high frequency energy and rejects low frequency energy
2. Mode (Common or Differential)
    - Common-mode noise injects current in the same direction in both the signal and return lines. Filter diverts common mode noise current to ground.
3. Amplitude (Surge suppression)
    - Amplitude selective filters reduce large transients or spikes e.g. surge suppressors.

- Time-average filter - Implemented in software, reduce the effect of noise on data within a signal.
- Time-synchronous filter - Stop running at periodic disturbance e.g. periodic switching in power supply.

Ferrite Beads

- Ferrite beads provide one form of filtering based on frequency.
- A magnetically permeable sleeve that fits around a wire. It presents inductive impedance to signals that attenuates high frequencies.
- Ferrite beads are best suited to filter low level signals and low current power feeds to circuit board.
- A ferrite bead is a passive electric component used to suppress high frequency noise in electronic circuits.
- It is a specific type of electronic choke. Ferrite beads employ the mechanism of high dissipation of high frequency currents in a ferrite to build high frequency noise suppression devices.
- The ferrite bead is effectively an inductor with a very small Q factor.
- For a simple ferrite ring, the wire is simply wrapped around the core through the centre typically 5 or 7 times. Clamp-on cores are also available, which can be attached without wrapping the wire at all.

Decoupling and Bypass Capacitors

- They provide Filtering based on frequency
- They filter and smooth out the spikes in DC power of Ics
- During a logic transition, a momentary short circuit from power to return in a digital device demands a large current transient. A decoupling capacitor can supply the momentary pulse of current and effectively decouple the switching spike from the power supply.
- They reduce the impedance of power supply circuit.
- Inductance in the power supply attenuates the effect of switching current transients by producing large voltage spikes.
- Decoupling capacitor provides this demand for shorter time.
- Mitigate the effect of inductance by reducing effective loop area between Power supply and the ICs.
- $Z\,0 = SQRT(R^2+[WL - (1/WC)]^2)$
- Reduce impedance of power supply
- If you arbitrarily make the decoupling capacitor too large, you will move the resonance frequency of the supply inductance and decoupling capacitor down into the range of operation of your circuit and cause excessive ringing in the supply.
- Also, large capacitors have larger parasitic inductances than smaller decoupling capacitors.

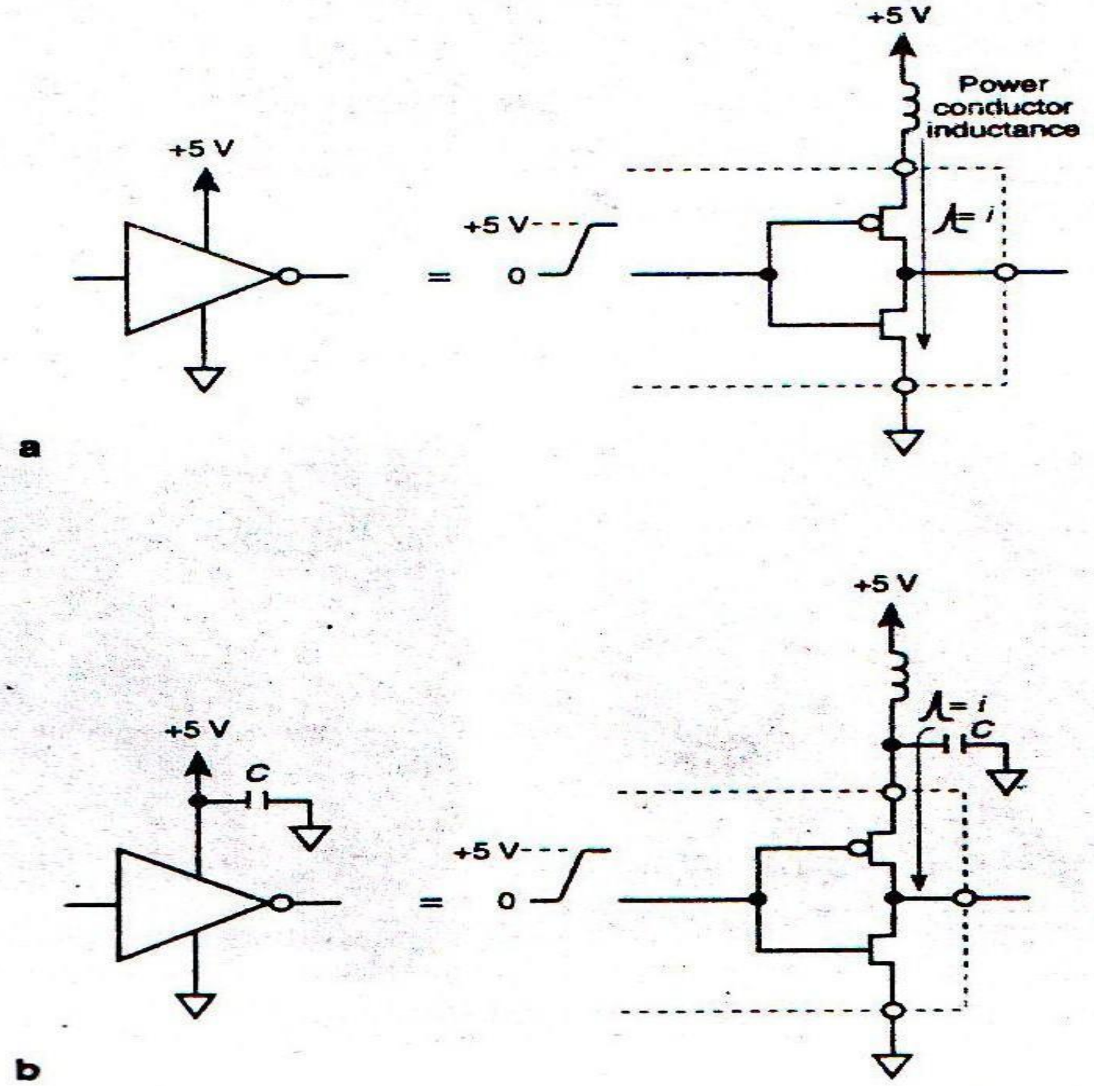
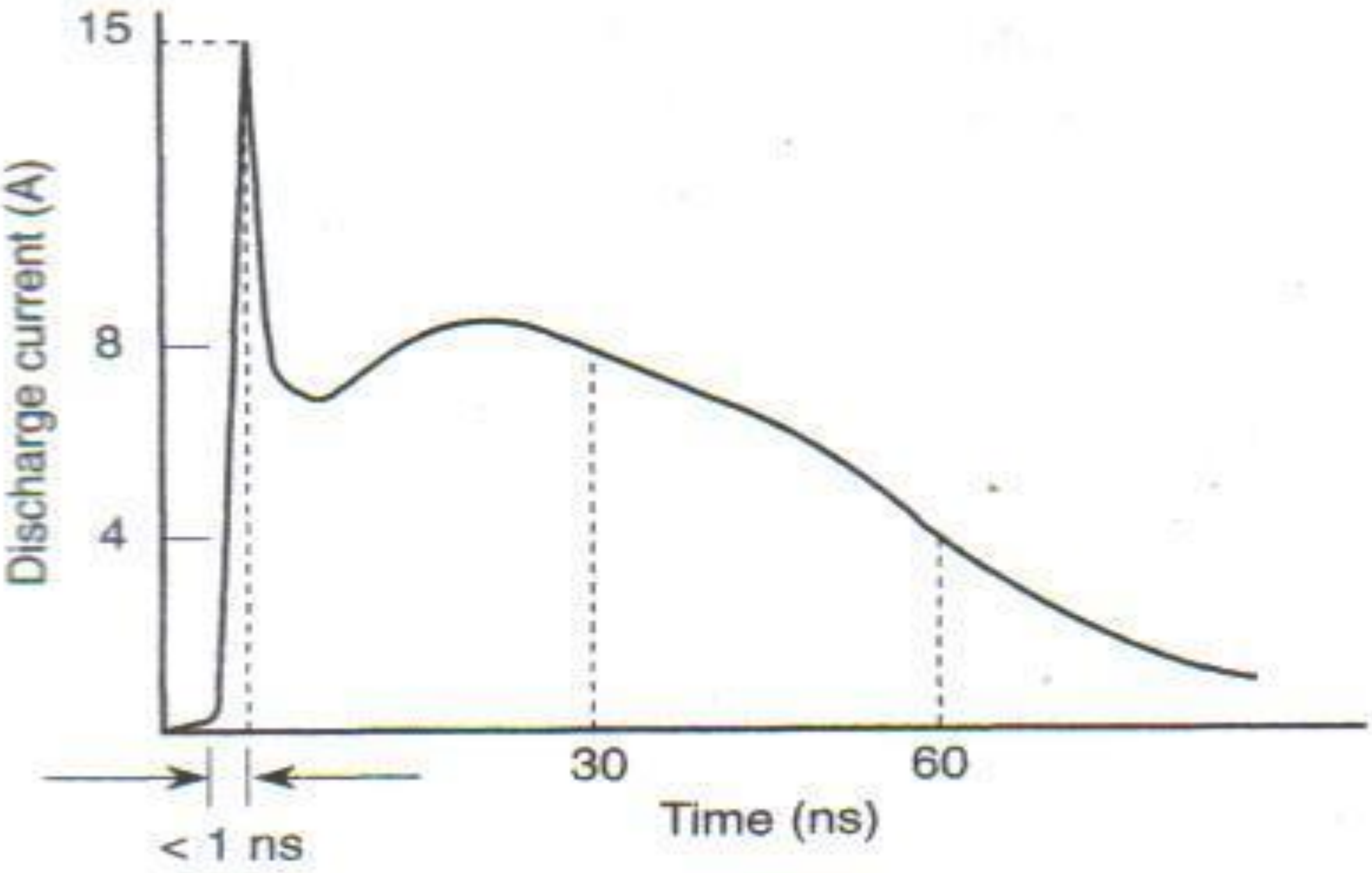


FIG. 6.29 Decoupling, or bypass, capacitors. They filter current transients when transistors switch within a digital logic gate. (a) As input voltage changes logic level, both transistors turn on, momentarily producing a short circuit and drawing a current pulse from the DC power source. (b) A decoupling capacitor can supply the current pulse and reduce the transient propagated through the DC power supply.

Electrostatics Discharge

- Electrostatic discharge (ESD) is a discharge at very high voltage and very low current that readily damages sensitive electronics.
- ESD can range from hundreds to tens of thousands of volts. Any instrument containing integrated circuits is susceptible.
- ESD transfers electrical charge in three stages: pickup, storage and discharge.
- Usually mechanical rubbing between dry, insulated materials transfers the charge from source to storage. Often the storage medium is person, who then unwillingly delivers the damaging discharge.
- Proximity or physical contact discharges the charge from storage.
- Several conditions including humidity, speed of the activity, and material affect the charge transfer.
- The discharge waveform of ESD has a fast rise time and short duration. The below figure illustrates sample waveform for simulating ESD while testing products.


15
8
4
Discharge current (A)
30
60
Time (ns)
< 1 ns

- For prevention, we need to eliminate the activities and materials that create high static charge control methods including the following.
  - Grounding
  - Protective handling
  - Protective material
  - Humidity
- Checklist to make work areas less prone to ESD
  - Use a “static-free” workstation, and wear a wrist ground strap
  - Discharge static before handling devices
  - Keep parts in original container
  - Minimize handling of components
  - Pickup devices by their bodies, not their leads
  - Never slide a semiconductor over any surface
  - Use conductive or antistatic containers for storage and transport of components
  - Clear all plastic, vinyl, Styrofoam from work area

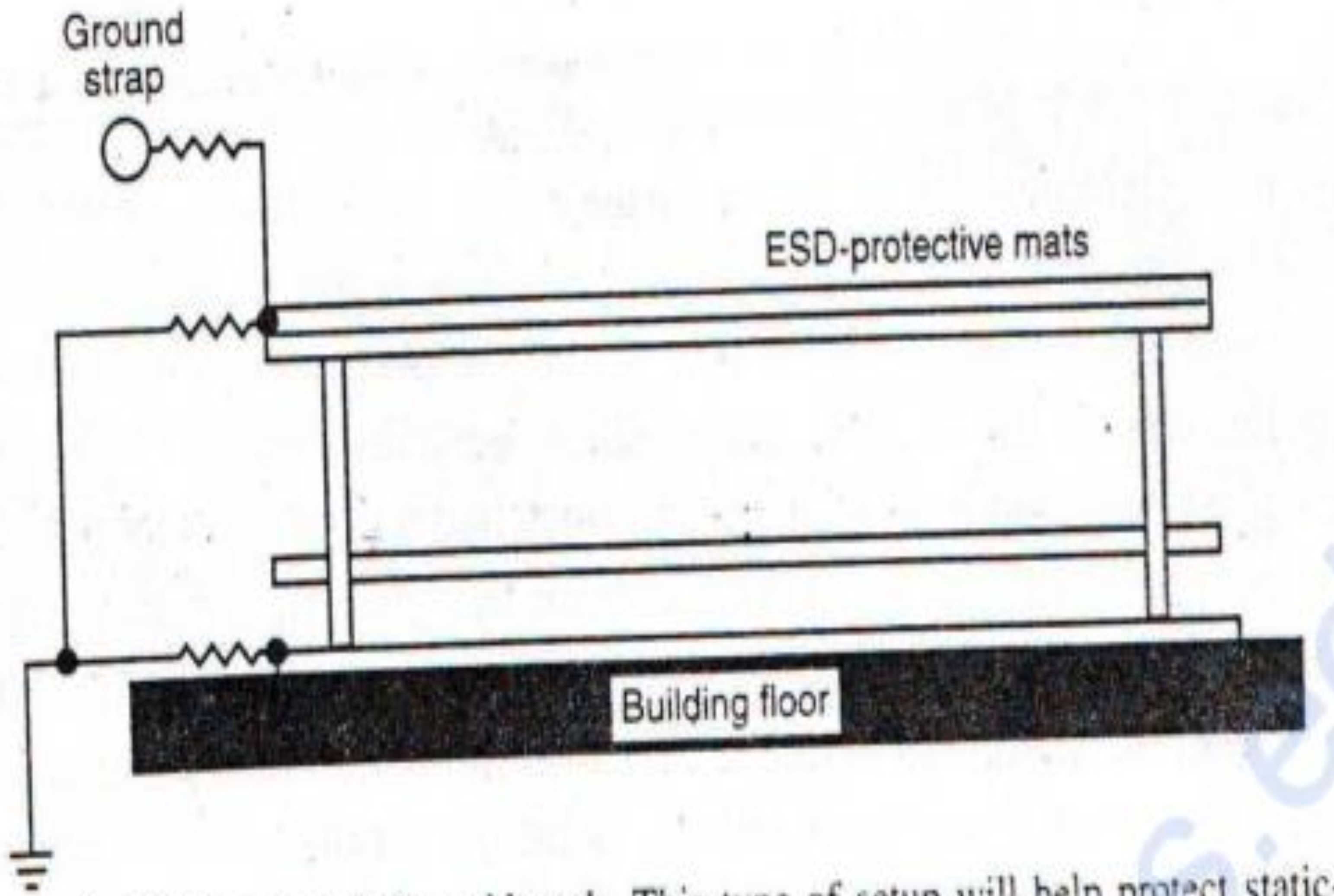


FIG. 6.43 Typical ESD-grounded workbench. This type of setup will help protect static-sensitive components.

From Symbols to Substance

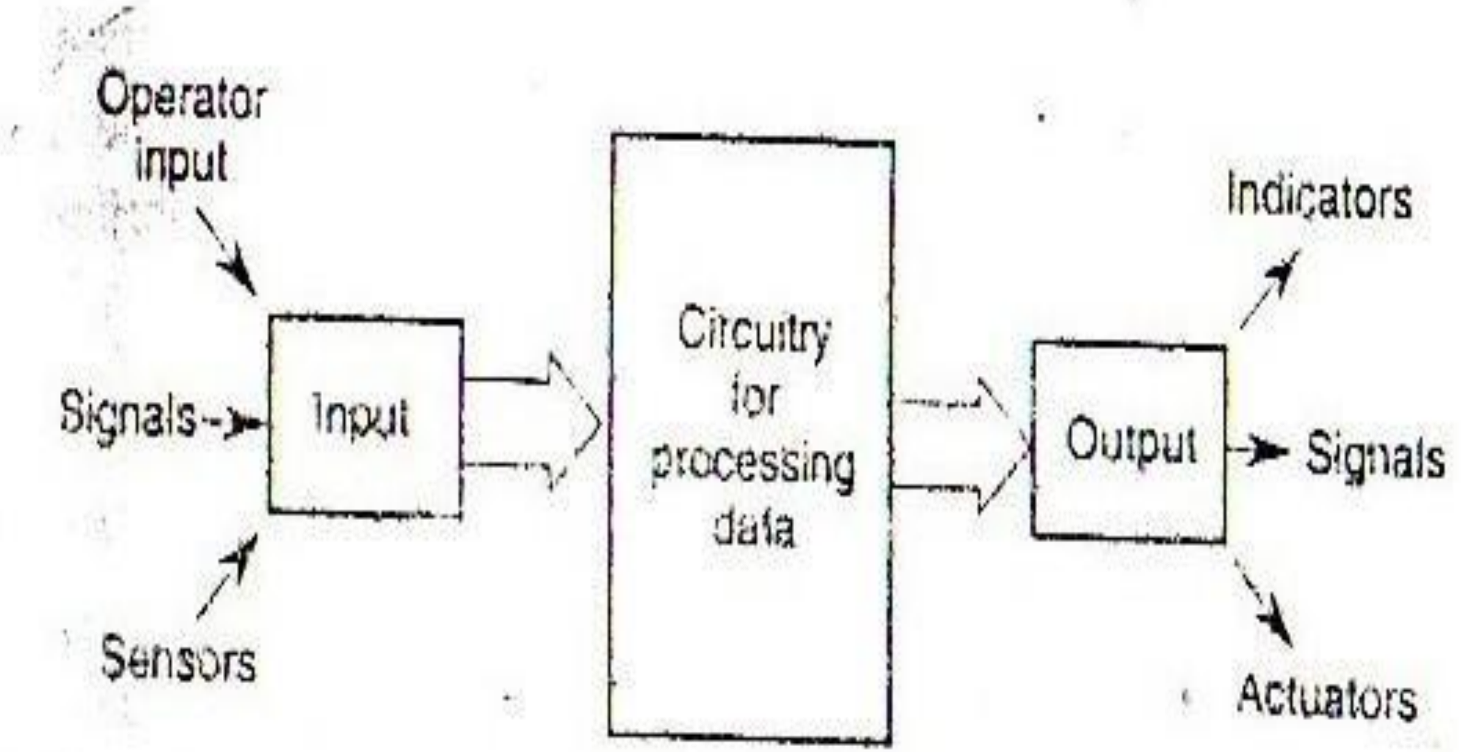


FIG. 7.1 General configuration for the circuits within a system.

Converting requirement into design

- Establishing requirement is the most difficult part of the circuit design.
- Experience is the best guide for setting requirements
- General to specific approach of establishing requirements:
  - Start by defining the desired function in broad term
  - Redefine the function with operational concerns
  - Settle on exact regulations and specification
- Setting specifications is one of the most difficult parts of engineering where good judgment and experience are necessary.
- Requirements often change late in the effort and spoil the design
- Some principles bound the design problem
  - E.g. use of electromagnetic spectrum
- Time and effort in design increases as the complexity of the function of system increases

- Choice of certain technology and devices are the result of good analysis and may depend on different factors
  - E.g. choice of a microprocessor for a system
  - Choice of A/D and D/A converters
- Technology drives the requirements. Audio frequency (~KHz) - Discrete components, wirewrap or PCB. Radio frequency (~MHz) - PCB (transmission line effect). Microwave frequency (~GHz) - RF design (Geometric structures).
- **Throughput**: The average rate of successful message delivers over a communication channel.
- Knowing region of operation, we can pick option available for circuit design. Right choice - Part count (↓), board space (↓), Power (↓), Cost (↓), time to market (↓), reliability (↑).

Table 7.1 Some requirements that drive design

System concern	Requirement	Parameters
Function	Response times	s, min, hr
	Data rates	Mbytes/s, kbits/s
	I/O drive	A, V
	Reliability—MTBF	hr
Regulations	FCC	
	UL	
	Military specifications	
Environment	Volume	in^3, m^3
	Weight (mass)	lb, kg
	Vibration	g, m/s^2
	Shock	g, m/s^2
Operation	Bandwidth	Hz
	Resolution	%, mV/LSB
	Speed	ns, μs, ms
	Accuracy	%
	Power consumption	mW
	Noise	$nV/\sqrt{HZ}$, SNR

Note: FCC = Federal Communications Commission; LSB = least significant bit; MTBF = mean time between failures; SNR = signal-to-noise ratio; UL = Underwriters Laboratories.

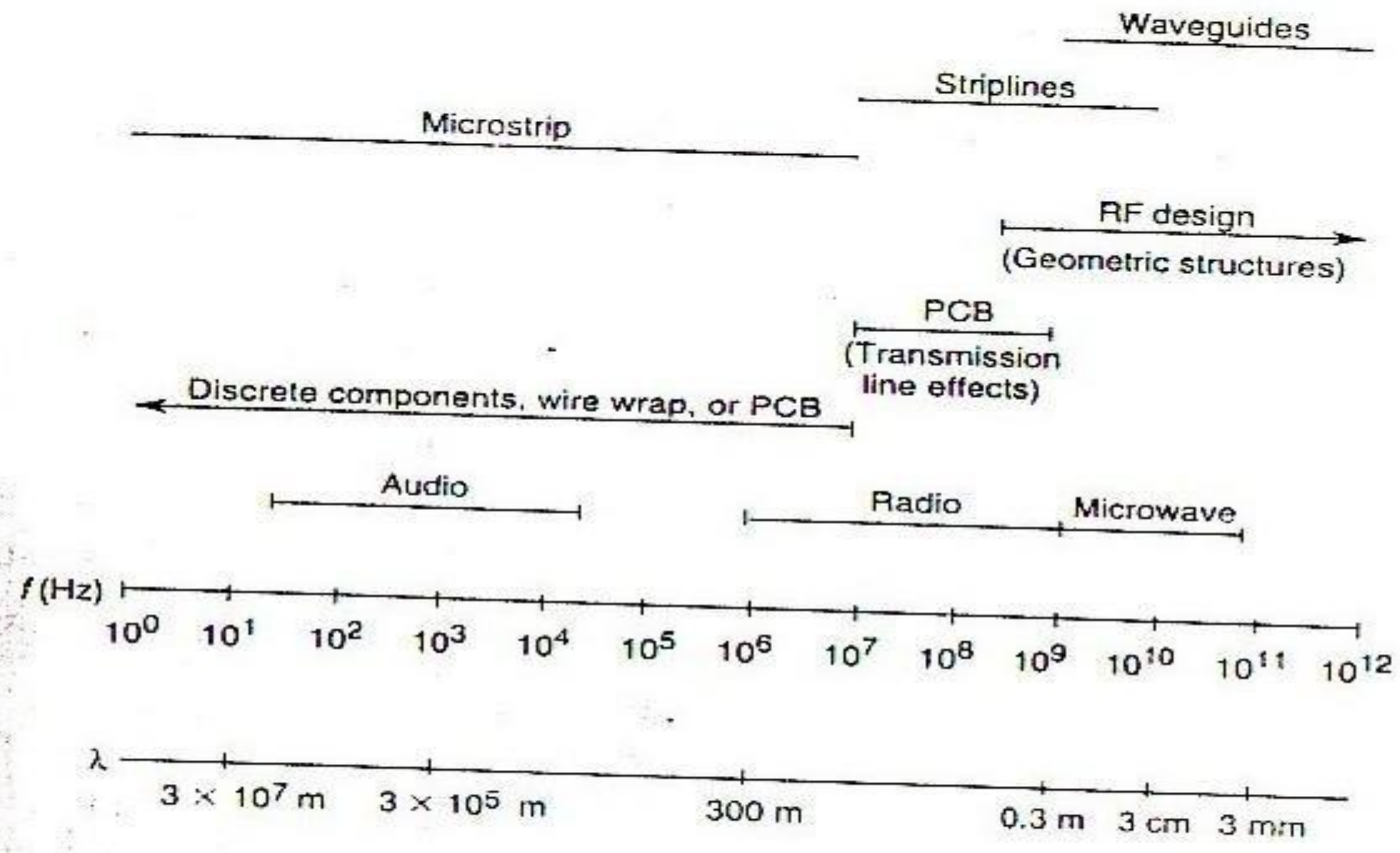


FIG. 7.2 A portion of the electromagnetic spectrum and corresponding technology applications.

Reliability

- How long the product will last?
- Two factors role in the reliability:-
  - Complexity:- Fewer part better
  - Design margin:- We must allow for stressing of components
- Two methods to measure reliability:-
  - Model prediction:- help update estimates of reliability but are limited and cannot predict every outcome
  - Prototype test:- find out many weaknesses and problems but are time consuming
- Combination of both is mostly used
- Standard methods for modeling use formulas based on practical experience of failure rates and physical knowledge to relate environmental factors to the reliability of electronic components.
- The failure rate for a component is a generally a base rate modified by various factors.

- The failure rate for a component is a generally a base rate modified by various factors
  $\lambda = \lambda_b \pi_e \pi_q \pi_a$ ……………………………(1)
  Where, λ = failure rate, λ_b = base failure rate
  π_e = environmental factor, π_q = quality factor
  π_a = acceleration factor
- Reliability of a component is a function of failure rate:
  $R(t) = e^{-\lambda t}$ ……………………….(2)
  Where $R(t)$ = Reliability, λ = failure rate, t = time

- Reliability of a System is a product of all component reliabilities:

  $$R_{System} = \prod_{i=1}^{n} Ri \qquad \text{……………………….(3)}$$

  Where R_{system} = reliability of the system, R_i = reliability of component
- Most failure rates relate acceleration factors (π_e, π_q, and π_a) to temperature, but not its applications.
- We may consider the application and some of the stresses and susceptibility factors might affect reliability
  - Corrosion, Thermal cracks, Electro migration, Secondary diffusion, Ionizing radiation, Vibration, High voltage breakdown, Ageing
- These can drastically alter reliability and still not predicted by standard models.

Fault Tolerance

- Goes beyond the design and analysis for reliable operation and reduces the possibility of dysfunction or damage from abnormal stresses and failures.
- Allows a measure of continued operation in the event of problem
- Three distinct area
  - Careful design
  - Testable function
  - Redundant Architecture

Careful Design

- Careful design can avoid many failures from abnormal stresses. Some design techniques that can reduce the probability of failure:
  - Reduce overstress from heat with cooling and low dissipation design.
  - Use optoisolation or transformer coupling to stop overvoltage and leakage current
  - Implement ESD (elctrosatic discharge) protection
  - Mount for shock and vibration
  - Tie down wires and cables
  - Prevent incorrect hookup; Use keyed connector

Testable Architecture

- The process of testing and diagnosing failures within a system.
- Two possible configurations of testable architecture:
  - **Simple Configuration:** Provides Probe points / test points for a technician or instrument to stimulate circuits and record responses. Only the trained personnel must disassemble the system and remove the circuit for testing.
  - **Complex Configuration:** Dedicated internal circuitry called **built in test** (**BIT**) that tests the system and diagnoses problems without disassembly of the equipment so adds complexity and reduces reliability. The trade off for BIT is quicker diagnoses and repair versus higher reliability.
- An appropriate calibration standard is always necessary when you measure a result.

Redundant Architecture

- The most complex and fault tolerant architecture are redundant architectures. They use multiple copies of circuitry and software to self check between functions.
- It is justified only when downtime for repair and maintenance cannot be tolerated.
- **Doubly redundant architecture:** merely indicates a failure in one of the subsystems; this allows for quick repair.
- **Triply redundant architecture:** uses voting between the outputs of three identical modules to select the correct value. It can have failure and still operate correctly.
- **Dissimilar redundancy:** compares the output from modules with different software and hardware to select the correct value. It can survive failure and even indicate errors in design if one system is coded correctly and the others are not.

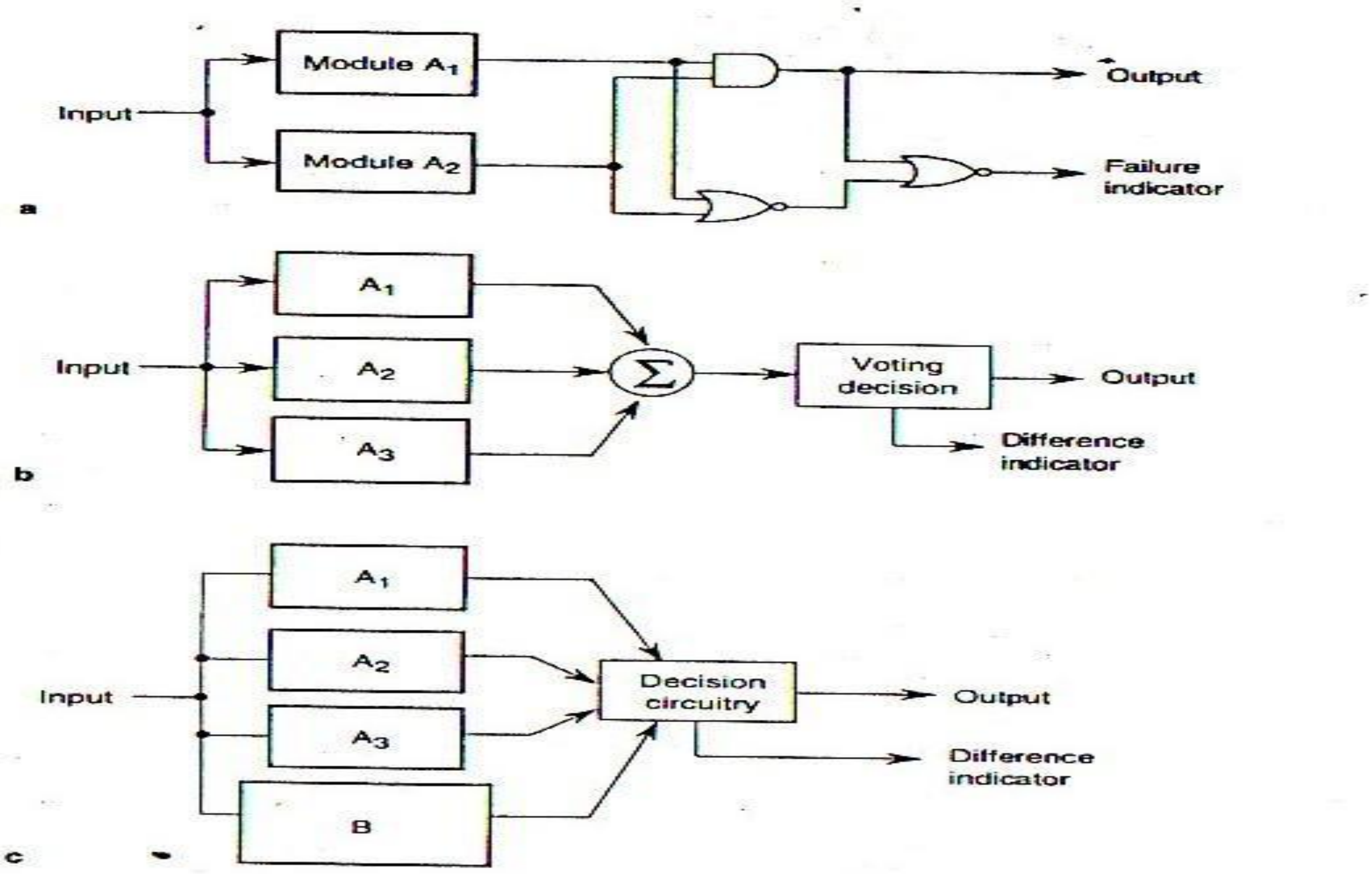


FIG. 7.6 Different configurations for redundant architecture. (a) Doubly redundant architecture ca signal failure. (b) Triply redundant architecture operates in spite of failure. (c) Dissimilar redundanc operates in spite of failure and indicates improper design or operation.

Bandwidth, Decoupling, Crosstalk, Impedance Matching

Bandwidth

- Limiting the bandwidth of the signals within a system is the most effective way to reduce noise, EMI and problems with transmission lines.
- May limit the bandwidth either by increasing the rise or fall times of the signal edges or by reducing the clock frequency.
- Selecting the appropriate logic family will set the edge rates and the consequent limit on transmission line concerns.
- One criterion for selecting logic according to transmission line effects is a ratio less than 4 between the rise time, t_r and the propagation delay, t_p i.e. (t_r/t_p <4).
- Slower edge rates allow longer interconnections between circuits.

Decoupling

- Switching of digital logic causes transients of current on the voltage supply through inductive impedance of the circuit
- Decoupling capacitor minimizes inductive loop area thus reducing impedance of power supply circuit.
- Shortest possible path for decoupling capacitor is best.

General recommendation for Decoupling:

- Use decoupling capacitor near each chip for two sided board
- Use a large filter capacitor at the power entry point
- Use a ferrite bead at the power entry point to the circuit board

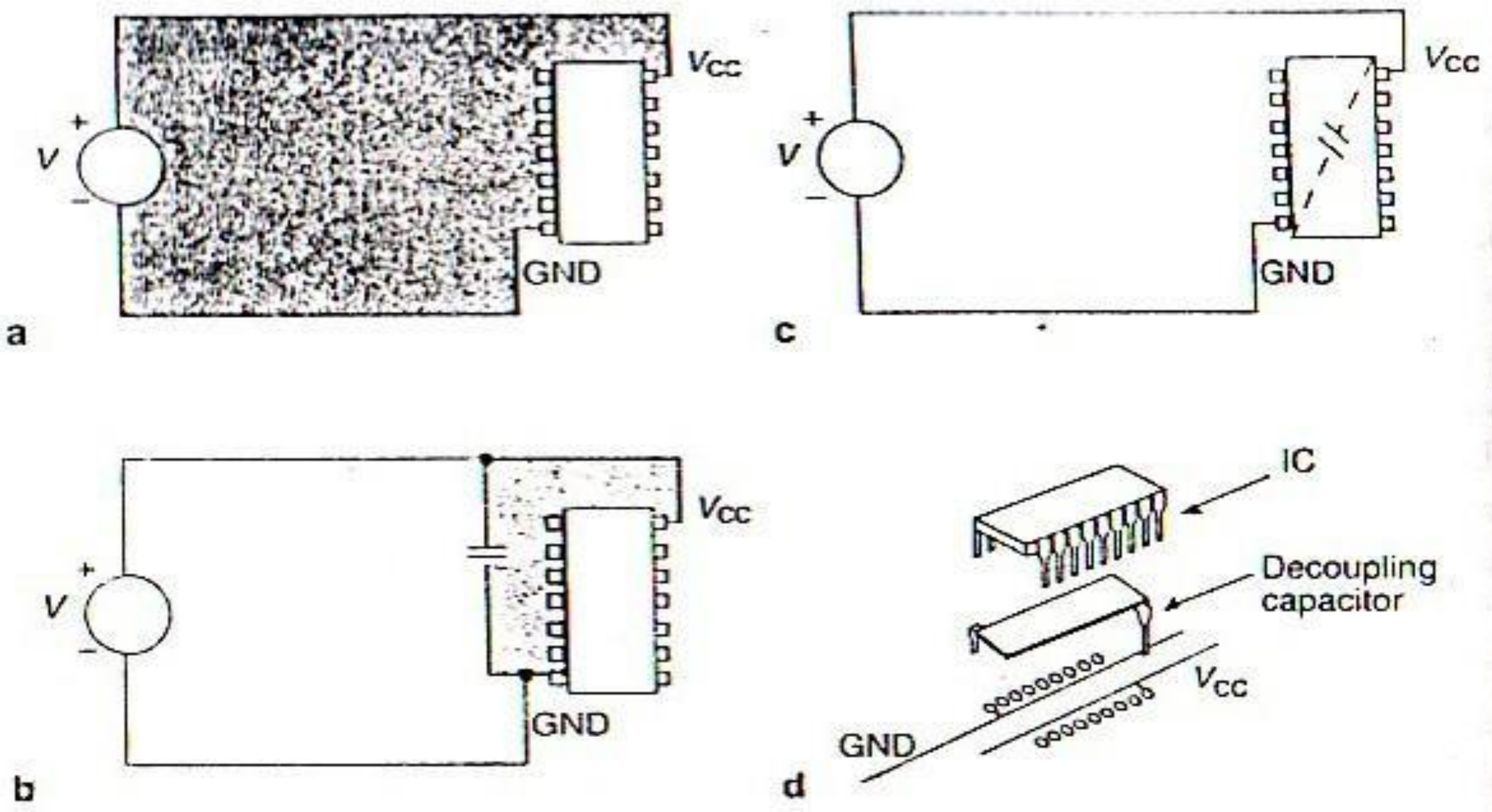


FIG. 7.8 Adding a decoupling capacitator to reduce the power-distribution impedance and consequentl noise: (a) Large inductive loop area without decoupling capacitor. (b) Decoupling capacitor reduce the inductive loop. (c) and (d) Closer to the chip is better.

Crosstalk

- Coupling electromagnetic energy from an active signal to a passive line
- Coupling mechanism:- capacitive or inductive
- Depends on line spacing, length and characteristic impedance, signal rise times
- **To reduce crosstalk:**
  - Decrease coupling length and characteristic impedance
  - Increase rise time of signal
  - Better layout and design of circuits
- **Avoiding Crosstalk**
  - Don’t run parallel traces for long distances – particularly asynchronous signal
  - Increase separation between conductors
  - Shield clock lines with ground strips
  - Reduce magnetic coupling by reducing the loop area of circuits
  - Sandwich signal lines between return planes
  - Isolate the clock, chip-select, chip-enable, read and write lines

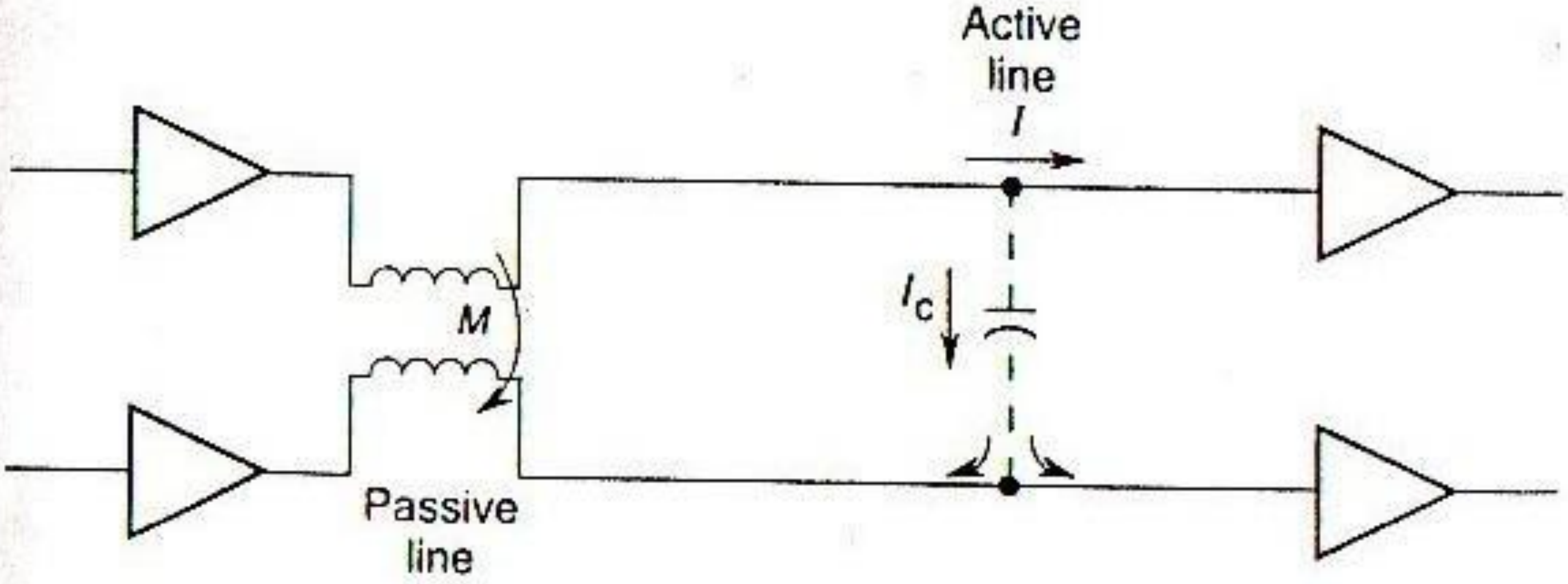


FIG. 7.10 Crosstalk—the inductive and capacitive coupling of signal from an active line to a passive line. (M = mutual inductance; I_c = capacitive current.)

Impedance Matching

- The reflection coefficient for a signal passing from medium 1 to medium 2 is given by: $\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$
- Where η_i is the intrinsic impedance of medium i and is given by

  $$\eta_i = \mathrm{sqrt}\ (\mu_i/\varepsilon_i)$$
- Reflection coefficient will be zero when $\eta_1 = \eta_2$
- Impedance matching makes the source and termination impedance equal to the characteristic impedance of the transmission line so that it will eliminate the reflections of signals that cause ringing (oscillations), undershoot, and overshoot in the signal pulses.
- Impedance discontinuities occur in two configurations endpoint and stub.
- **End point discontinuity: -** the ends of the transmission line don't match its characteristic impedance of the transmission line.
  - Add series resistances at the end until the total impedance equals the line impedance.
  - Terminate the other end of the signal line from driver.
- **Stub discontinuities** cause impedance mismatch and signal reflection by connecting multiple circuits to a single line.
  - Each Connection of a stub divides the impedance and splits the power of the signal
  - Make them very short, even zero to reduce the effect of stub discontinuities
  - Good layout and design

PCB (Printed Circuit Board)

- Etched and plated connections
- Make automated placement and soldering of components possible
- Control impedances more effectively than wire-wrap
- Cost effective, manufacturing edge and reliability
- **Single Sided PCB**
  - Low frequency operation (< 25 KHz)
  - Signal cross over wire jumpers are used
- **Double Sided PCB**
  - Signal traces on both sides of the circuit board and plated through vias.
  - Can support higher frequency operation if laid out very carefully.

- **Multilayer PCB**
  - A stack of alternate layers of copper-clad laminate or core and prepreg.
  - 20 to 30 conducting layers laminated together
  - Control impedance much more tightly and are absolutely necessary for high frequency circuits
  - Through-hole vias penetrate all layers and can connect signals on each layer.
  - Buried vias connect traces on two sides of an internal layer
  - Blind vias are exposed on one external layer and connect traces on the two sides of that layer.

MCM (Multi-Chip Module)

- Higher level of circuit density by bonding the base die of ICs onto a substrate
- Compact packing improves signal speeds and reduces load capacitance
- Expensive to fabricate

Component Placement

- Affects circuit operation, manufacturing edge, and the probability of design errors.
- General rules:

  1) Group high current circuit near the connector to isolate stray currents and near the edge of the PCB to remove heat.

  2) Group high frequency circuits near the connector to reduce path length, crosstalk and noise.

  3) Group low power and low frequency circuits away from high current and high frequency circuits

  4) Group analog circuits separately from digital circuits.
- Grouping components and circuits appropriately will reduce crosstalk and noise and will dissipate heat efficiently.

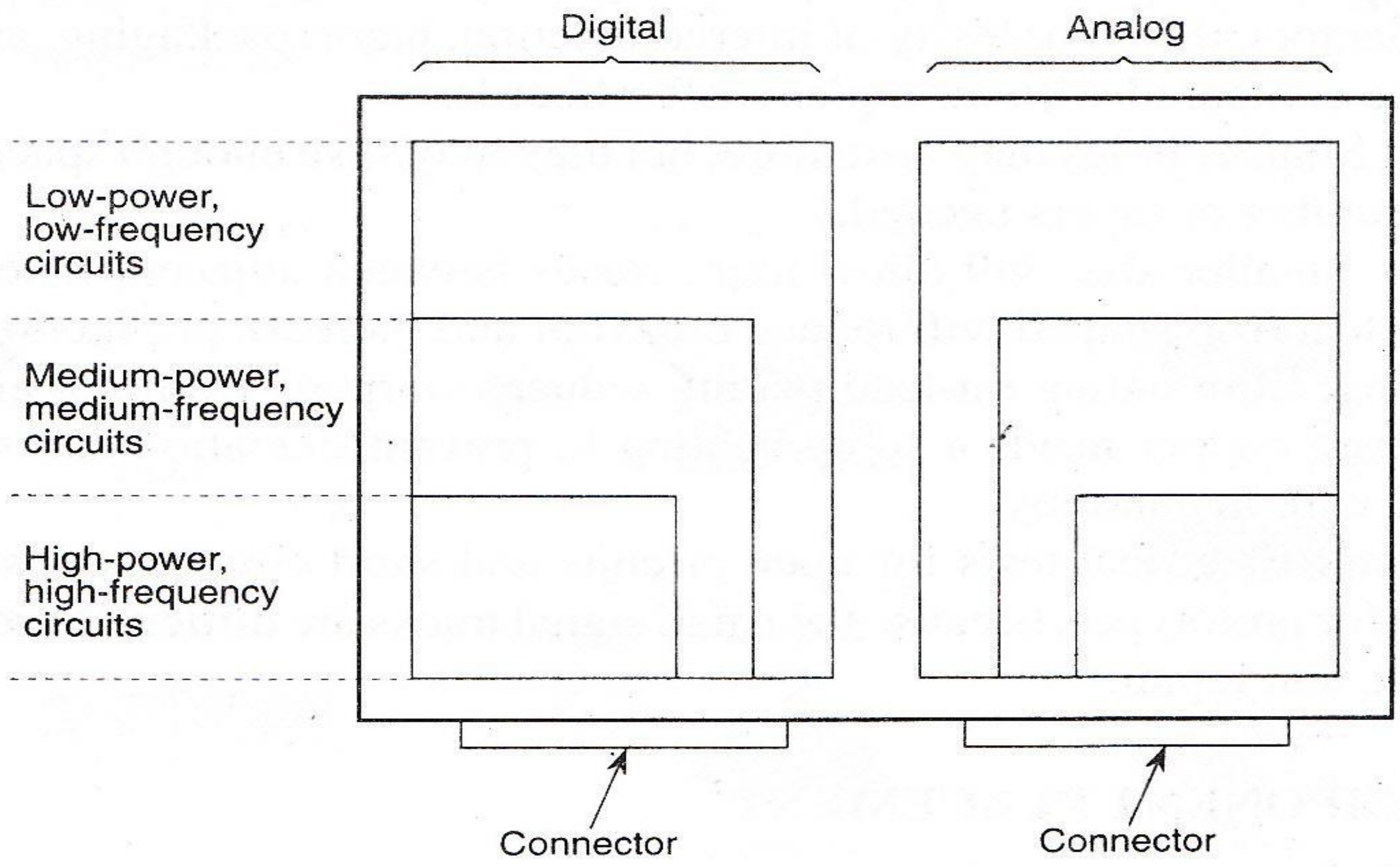


FIG. 8.7 Grouping of components. Arrange high-power and high-frequency circuits near the connector and away from the low-power circuits.

Routing Signal Traces

- Poor layout - false triggering of logic
- Due to formation of capacitive and inductive parasites with stubs, vias, IC pins, multiple loads and traces; setup and hold time violation and transmission delay.

Trace Density

- Trade off between greater cost and difficulty in producing the denser circuit board.
- As you squeeze signal traces together on a board, you can space components closer and reduce the size of the circuit boards.
- Smaller boards, allowed by higher trace densities, provide flexibility in packaging your product, reduce the cost of material and may degrade signal integrity.

Common Impedance

- Minimize the number of circuits that share the same return path. Voltage drops (caused by current switching) on the ground line (return path) increase system noise.
- Common impedance paths cause components to reside at different ground potentials from one another.
- You can reduce the voltage drops, and hence the noise by lowering effective impedance
- Unbroken return plane is the best way.
- Choosing the right logic family and using decoupling capacitors will help by reducing the magnitude of current pulses.

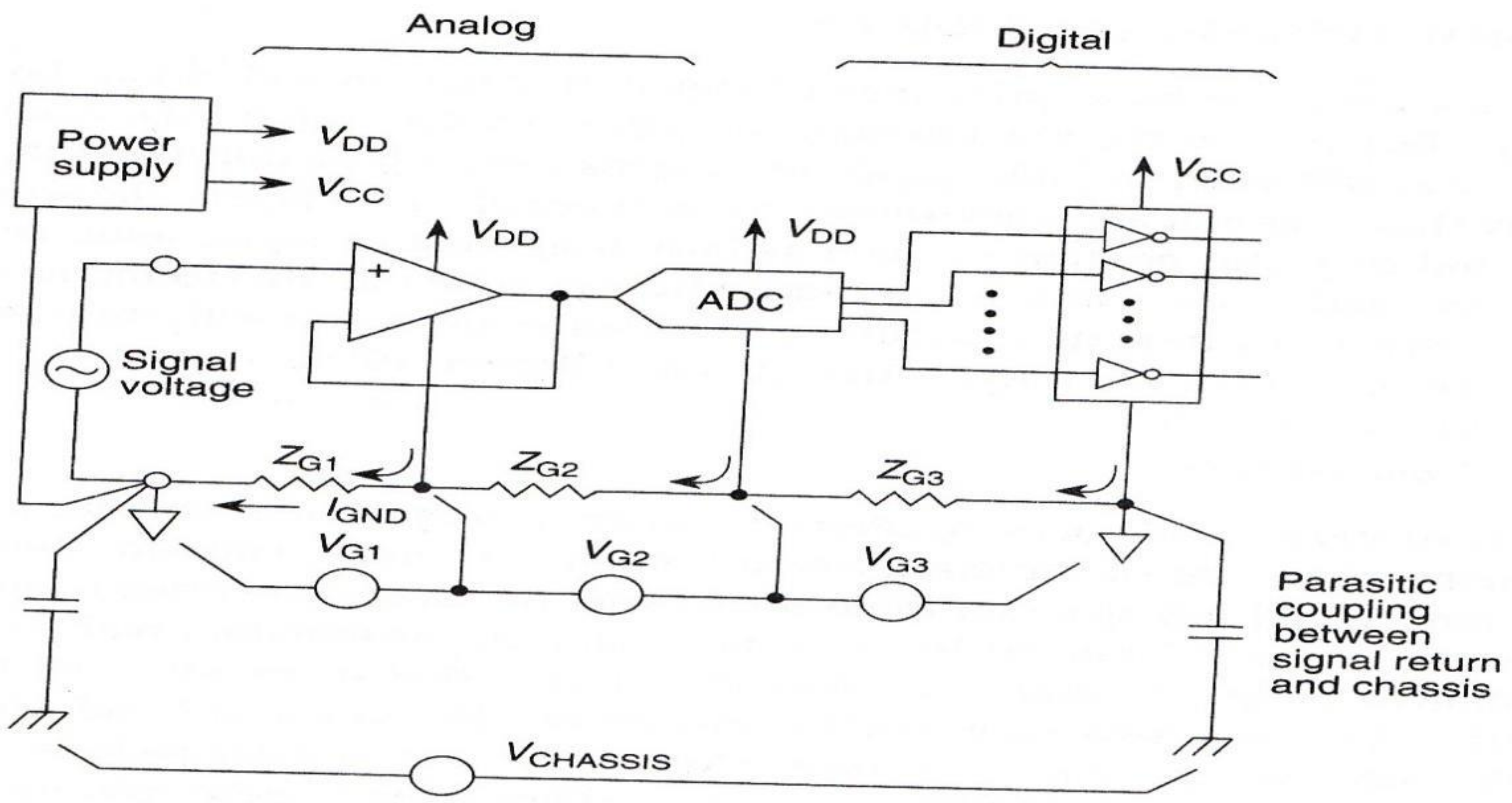


Fig. 8.8 Common impedance. Common-impedance paths cause components to reside at different ground potentials from one another. Larger common impedance will cause larger ground potentials (V_{G1}, V_{G2}, V_{G3}) and coupling to the chassis ($V_{CHASSIS}$); these will bias the signal voltage and introduce error.

Distribute Signal and Return

Distribute signal and return

- Address the issues of return path early in design.
- Long return path can shift the ground potential excessively, decrease noise margins, and cause false switching.
- If the return is longer than signal, then the current has high inductance path that cause noise spikes in the ground system.
- Large loops of current have high inductance or impedance and radiated noise is often proportional to return path impedance and loop area.

Trace Impedance and Impedance Matching

- Impedance of signal conductors directly affects circuit operation.
- Low characteristic impedance (Z_0) radiates less and is less susceptible to interference than a circuit with higher impedance.
- Impedance mismatches lead to reflections that can both delay switching and trigger logic falsely.
- Reflection Coefficient = $(Z_L – Z_0) / (Z_L + Z_0)$
- Multiple loads have stubs, non-uniform impedance and mismatches that compromise noise margins.

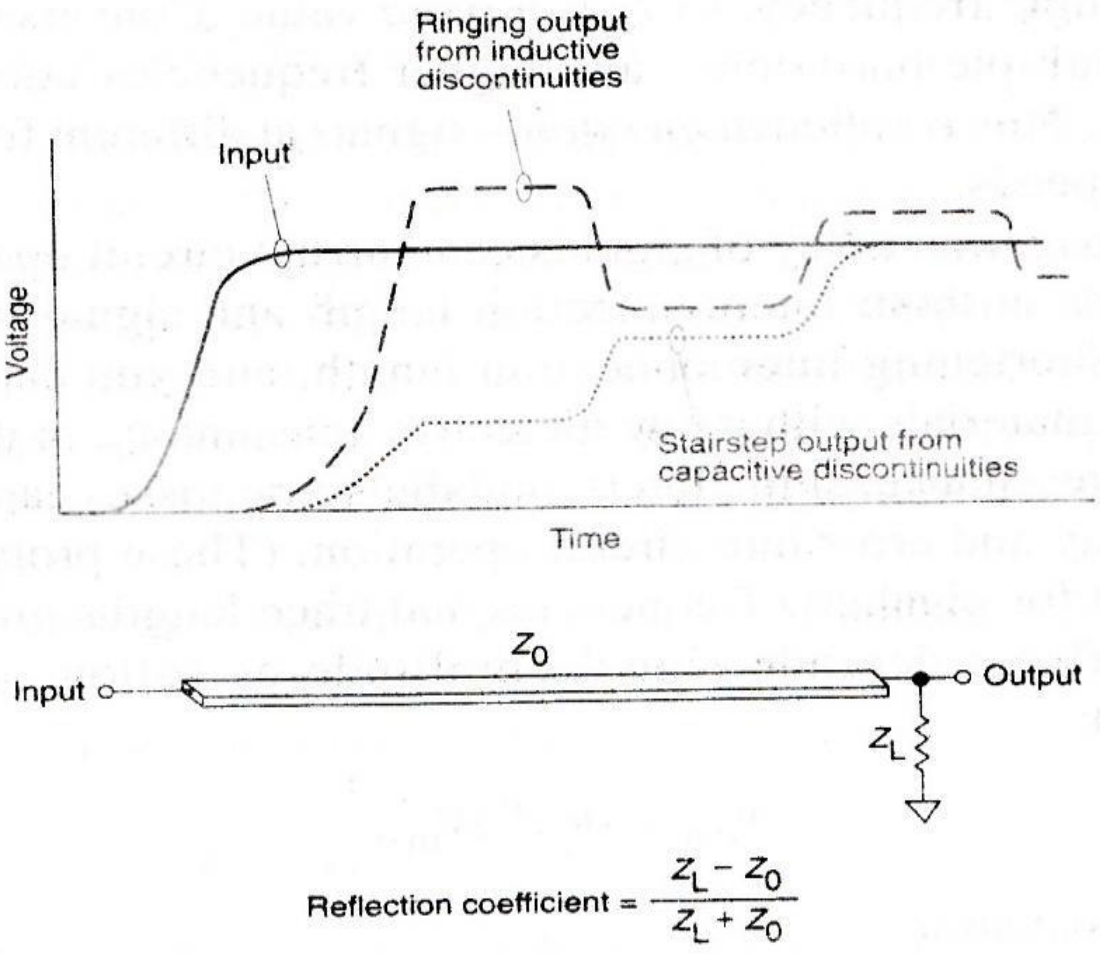


FIG. 8.10 Impedance mismatch and resulting degradation of signals along a printed circuit trace

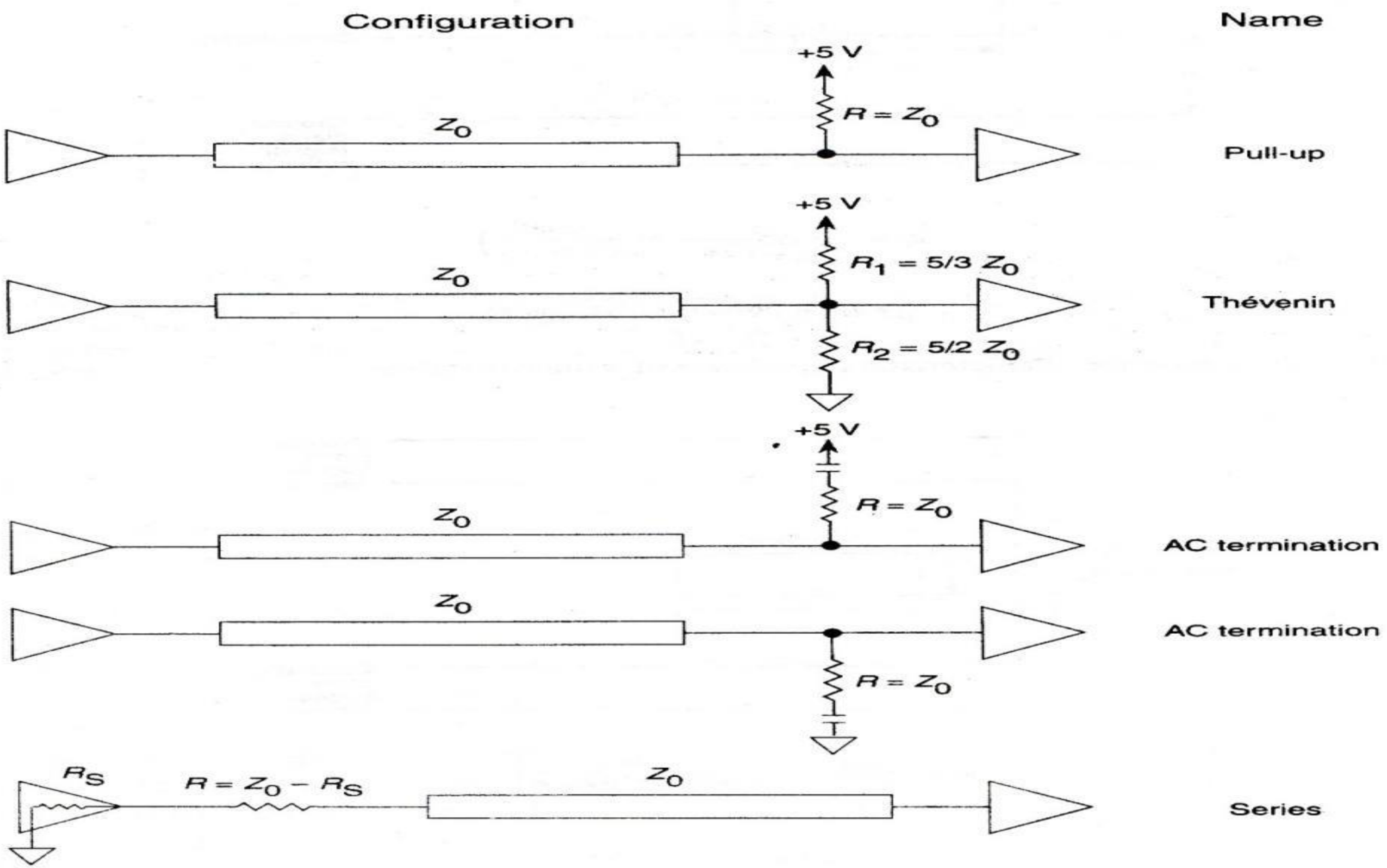


FIG. 8.11 A number of ways to terminate a signal trace.

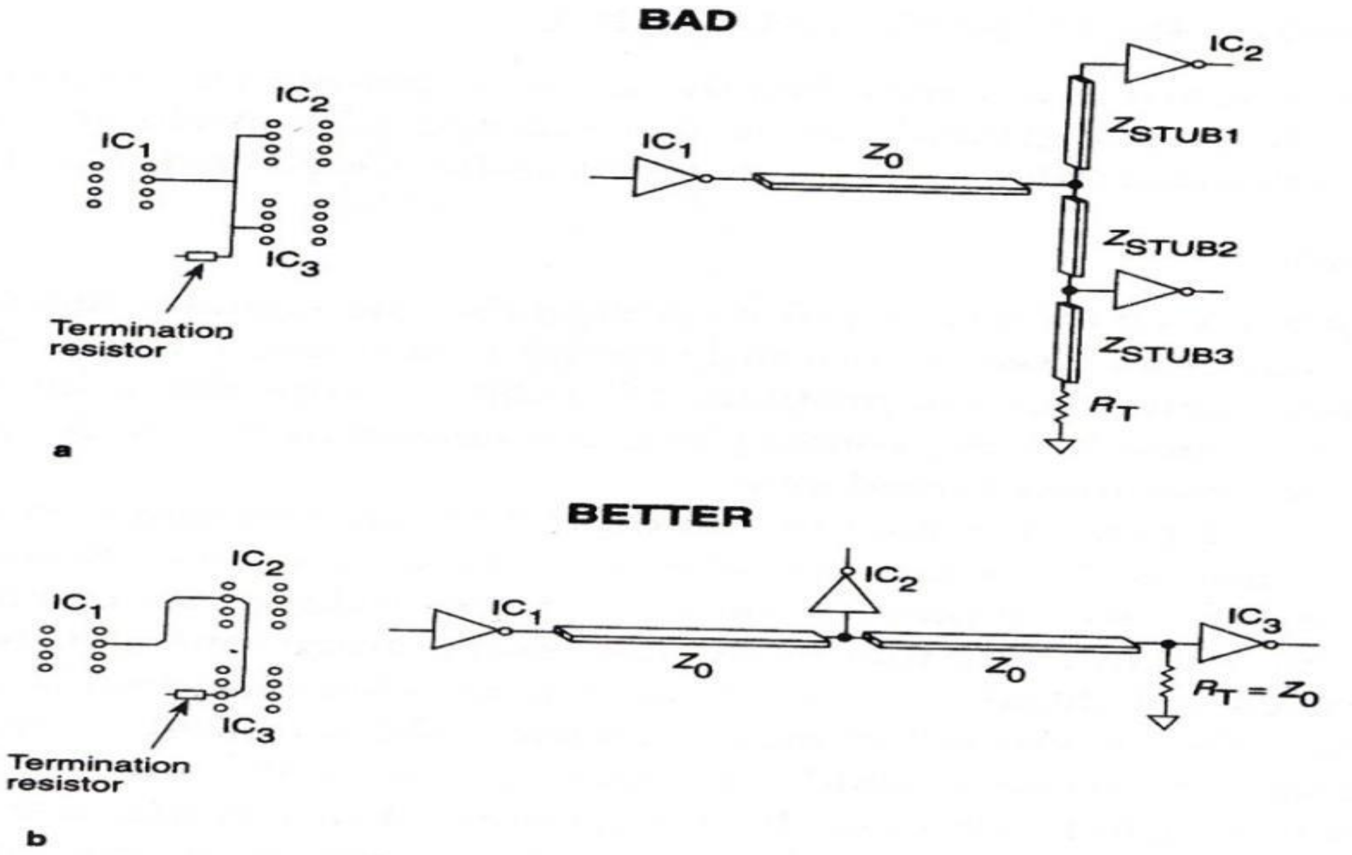


FIG. 8.14 Using a single serpentine trace to connect multiple inputs and eliminate stubs that cause reflections in the signal. (a) Poor layout degrades signal edges at the inputs to IC_2 and IC_3. (b) Using both a serpentine trace and a termination resistor near the last input eliminates stubs and preserves signal integrity.